Arunabha Majumder

Robotics and Mechatronics Engineer | Hands-on Integration, Sensors and Actuators, Testing

+45 71 62 24 00 | Aalborg, Denmark (open to relocating to Paris area)
somrkmv1997@gmail.com | 97arunabhasit027@gmail.com | Portfolio: arunabha22.github.io/arunabha-majumder

SUMMARY

Robotics and mechatronics engineer who works best with hands on the hardware. During my PhD at Aalborg University I integrated motors, encoders and force sensors into robotic devices, built and debugged my own test rigs, and found root causes when things failed. I write clear technical notes and specifications, and I learn new tools quickly.

EXPERIENCE

PhD Researcher, Mechatronics and Robotics (project VIEXO) | 2023 - Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

- Integrated the motor, encoder and force sensors into a shoulder exoskeleton and built the test rigs around it; found and fixed faults in mounting, wiring, sensors and data logging.
- Designed the hybrid spring and 6 Nm motor actuator in SolidWorks and iterated it with 3D printing; it cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Validated a force estimate on a 5-bar parallel robot against a reference force sensor, within 10% RMS error.
- Worked in a team of 3 on an Assist-as-Needed controller (my part: equations such as the Fourier fit, and GUI behaviour); collected data from 12 participants with attention to safety at every session.
- Taught and supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Artificial Muscles (Master's thesis) | 2020 - 2022
CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated and assembled pneumatic muscles to an existing lab design, ran experiments on the lab's setup with valves, pressure sensors and DAQ, and used the data to design the outer PID loop of a gain-scheduled controller (0.3-0.78% error, IEEE CONECCT 2022).

Student Builds, Siddaganga Institute of Technology | 2017 - 2019
Tumakuru, India

- Designed, machined, assembled and tested a rehabilitation exoskeleton (Best Major Project Award, TATA Technologies grant); built a Go-Kart and an Efficycle hybrid vehicle.

SKILLS

- **Sensors and actuators:** Motors, encoders, force/torque and pressure sensors, pneumatic valves, DAQ, Arduino
- **Mechanical:** SolidWorks (CSWA), 3D printing, test rigs; machining and assembly (Bachelor's)
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic)
- **Drawing:** Diploma in Fine Arts; clear sketches and figures for specifications and reports
- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (defence expected in 2027) | 2023 - Present
Aalborg University, Denmark

Study abroad, National University of Singapore (NUS), Sep - Nov 2025

MTech, Mechatronics | 2020 - 2022
AcSIR - CSIR-CMERI, India

BTech, Mechanical Engineering | 2015 - 2019
Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. ISRM 2026 (Springer), hybrid actuated shoulder exoskeleton; IEEE CONECCT 2022.

REFERENCES

Available upon request.